From collisions to recoveries: MARSALA random access

CESE4055 Ad Hoc and Sensor Networks - Group Project 2

The question

How much can a receiver recover from a shared satellite return link when several users transmit in the same time slot? We compare three increasingly capable receivers: slotted ALOHA, CRDSA, and CRDSA with MARSALA. The central tradeoff is that packet repetition creates both *diversity* (more chances to observe a packet) and *contention* (more transmissions sharing the channel). CRDSA uses the diversity to cancel interference; MARSALA also combines collided replicas to unlock cases where cancellation alone stalls. The original CRDSA proposal is due to Casini, De Gaudenzi, and del Rio Herrero [1]; the MARSALA receiver method is described by Bui, Zidane, Lacan, and Boucheret [2].

Three ways to receive the same traffic

Receiver	Transmitter behavior	What the receiver can recover	Main limitation
Slotted ALOHA	One packet in one random slot	A packet alone in its slot	Every collision is lost
CRDSA	Several replicas in different slots	Singletons, then packets exposed by successive interference cancellation (SIC)	SIC stops when no singleton remains
MARSALA	Same transmissions as CRDSA	CRDSA recoveries plus packets decoded by localizing and combining collided replicas	More receiver work and dependence on signal quality

For a packet arrival load of `U = users / slots`, ideal slotted ALOHA has throughput approximately `U exp(-U)` packets per slot. It peaks at `1/e`, about 0.368. In a finite 100-slot frame, the exact singleton probability for one tagged packet is `(1 - 1/100)^(users - 1)`. With 75 users, this predicts about 0.355 delivered packets per slot, close to the measured 0.356.

CRDSA sends each packet in `R` distinct slots. A singleton is decoded, its other replicas are cancelled, and a collision may become a new singleton. This iterative process is often called *peeling*. A group of packets can form an interference loop with no exposed singleton; then peeling deadlocks. In the fixed example frame, slotted ALOHA decodes 31 of 75 packets, CRDSA with two replicas decodes 62, and MARSALA decodes all 75. The extra 13 are the packets that combining makes recoverable after CRDSA stalls.

MARSALA correlates collided slot signals to find replicas of one packet, phase-aligns them, and combines their energy before testing whether decoding is possible. For equal-power replicas and independent interference, a useful approximation is `combined SINR ~= R*rho / (1 + (k - 1)*rho)`, where `rho` is the per-replica signal-to-noise ratio and `k` is the collision depth. More replicas or higher SNR help; deeper collisions and a stricter required SINR hurt. This is an explanation of the mechanism, not an exact link budget.

What changes when parameters move?

The controlled study holds packet demand at **75 users per 100-slot frame**. Each point averages 60 frames. This equal-user comparison matters: the replica occupancy is `G = U*R`, so holding `G` fixed while changing `R` would change the number of users. The default is `R=2`, per-replica SNR `-2.2 dB`, and required SINR `0.61`. The accompanying parameter figure plots all three receivers; the following observations come from `parameter_study.csv`.

Replica count. With one copy, all three methods reduce to the singleton baseline, about 0.356 packets per slot. At two copies, CRDSA reaches 0.502 and MARSALA 0.750. At three, CRDSA reaches 0.631 and MARSALA stays at the 0.750 demand ceiling. At four, CRDSA falls to 0.428 because the extra copies make slots more crowded; MARSALA remains at the ceiling in this idealized model. Thus repetition is not free. Once all offered packets are delivered, further copies consume channel capacity without improving throughput.

SNR. For two replicas, increasing per-replica SNR from `-6` to `-4` to `-2.2 dB` raises MARSALA throughput from 0.502 to 0.633 to 0.750. ALOHA and CRDSA are almost unchanged here: the simulated singleton decoder has a large processing gain and already succeeds. The SNR change mainly decides whether a *combined collision* crosses the decoding threshold. Above the threshold, further SNR yields little gain because all 75 packets are already recovered.

Required SINR. At the default SNR, increasing the required SINR from `0.61` to `0.8` to `1.0` lowers MARSALA throughput from 0.750 to 0.640 to 0.502. At 1.0, combining adds no recoveries over CRDSA in this study. A stricter threshold represents a less capable code or a stronger reliability requirement. It improves neither throughput nor delay here; it only rejects borderline combined packets.

Offered load. As user demand grows, singleton slots become rarer, so ALOHA losses rise. SIC keeps CRDSA useful beyond ALOHA's peak, and MARSALA can unlock additional interference loops. Eventually deeper collisions reduce combined SINR and demand exceeds what the receiver can recover. The load plots show throughput, packet loss, and delay. Their horizontal axis is *replica* load `G`, not packet arrival load `U`; compare different `R` values at equal `U` using the controlled study above.

Throughput, loss, delay, and receiver cost

Throughput is delivered packets divided by frame slots. Loss is the fraction of offered packets not decoded in that frame. In the delay experiment, undelivered packets retry in new slots next frame and are dropped after 20 attempts. The plotted mean delay includes only packets eventually delivered; loss must be read beside it. At replica load `G=2.5`, CRDSA with two replicas has mean delay 4.62 frames, while MARSALA has 1.04 in this model. These figures illustrate how an unlocked SIC cascade shortens retransmission time.

MARSALA pays for that gain at the receiver. At each deadlock it tests correlations between candidate collided slots, then estimates and combines replicas. Search effort grows rapidly at heavy load: with two replicas and `G=2.5`, this implementation averages about 11,353 correlations per frame. This is a serious complexity cost, not a free throughput gain. It also shows why candidate selection and search limits matter in a practical receiver.

Scope of the evidence

The model uses 100 slots, 256 complex samples per slot, random phases, amplitude variation, and additive Gaussian noise. Decoding uses a threshold in place of a complete error-correcting code. More importantly, the receiver uses the generator's true occupancy, packet symbols, and amplitudes for its decisions; it does not perform blind packet identification or physical signal subtraction. It is an **oracle-assisted concept simulation**. The plots show the direction of the mechanisms and the interaction of controls, not validated satellite-link performance. In particular, the very high two-replica MARSALA throughput at heavy load should not be used as a real-world prediction. The live page demonstrates one frame at a time; the report figures average multiple frames to reduce dependence on the seed.

Conclusion

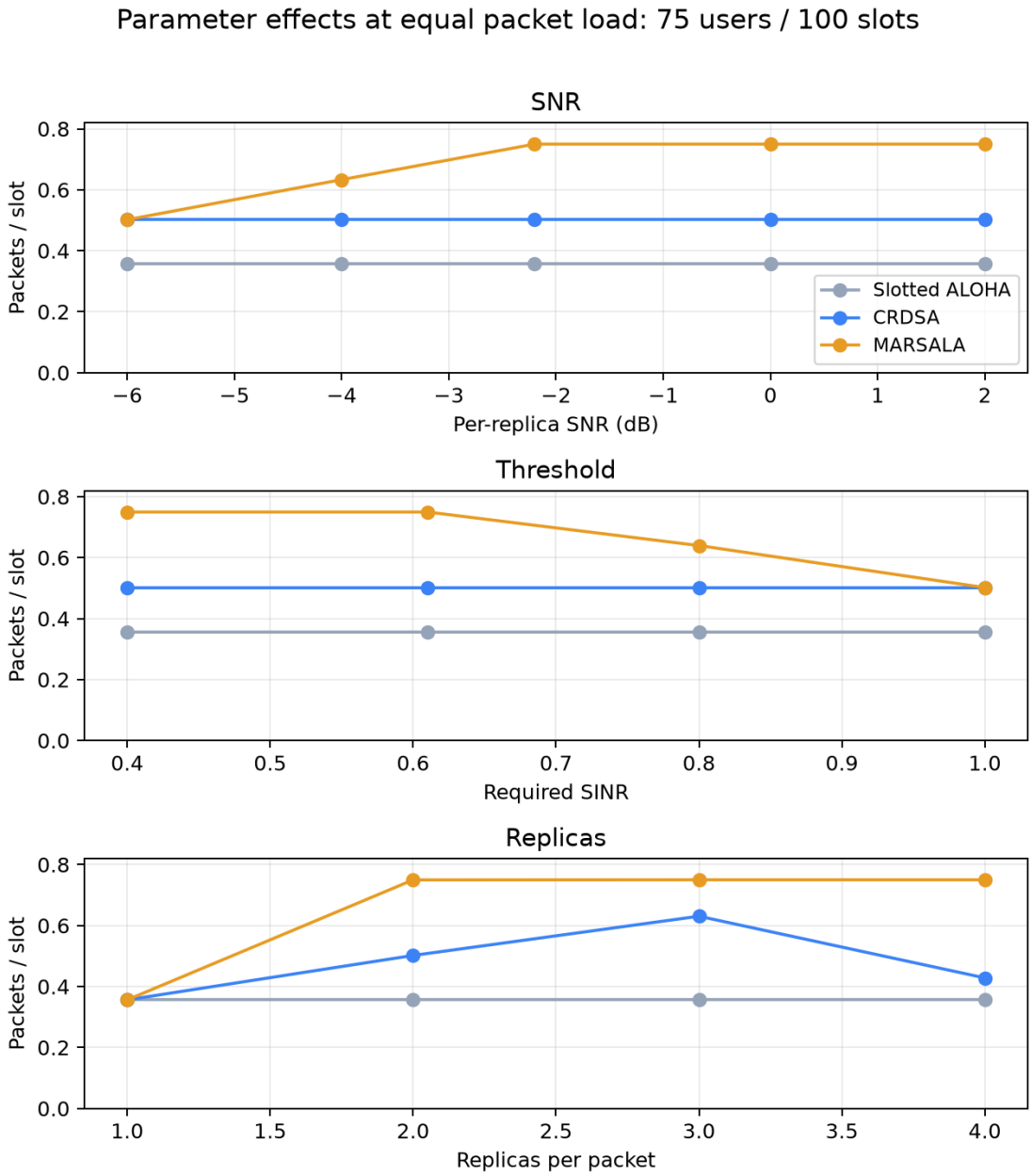
Slotted ALOHA is simple but throws away collisions. CRDSA turns some collisions into recoveries through repetition and SIC, until an interference loop blocks further progress. MARSALA can combine collided replicas to restart SIC, especially when SNR is adequate and the decoding threshold is reachable. Replica count and signal quality must be chosen together: more copies can help recovery but also occupy more slots, while stricter decoding and deeper collisions make combining less effective. The cost is greater receiver complexity and sensitivity to assumptions about channel knowledge.

References

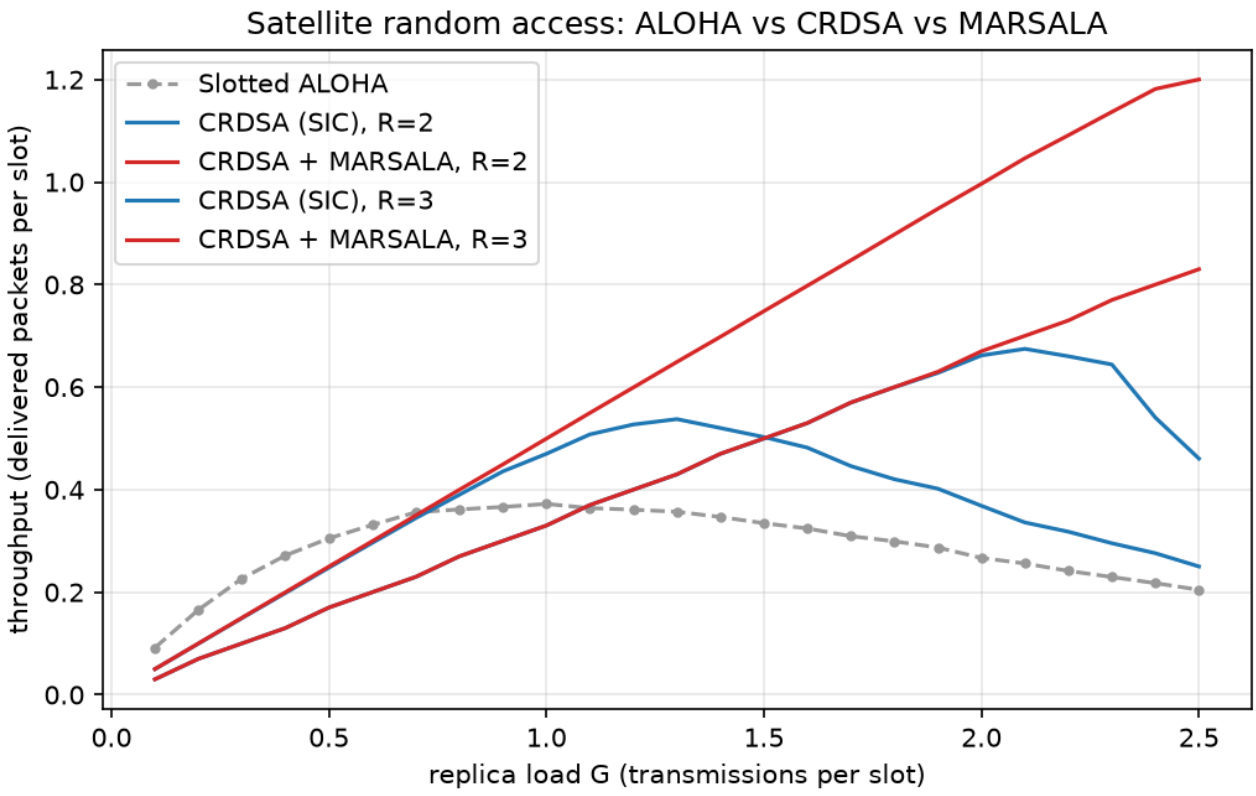
[1] E. Casini, R. De Gaudenzi, and O. del Rio Herrero, “Contention Resolution Diversity Slotted ALOHA (CRDSA): An Enhanced Random Access Scheme for Satellite Access Packet Networks,” *IEEE Transactions on Wireless Communications*, 6(4), 1408-1419, 2007. doi:10.1109/TWC.2007.348337

[2] H.-C. Bui, K. Zidane, J. Lacan, and M.-L. Boucheret, “A Multi-Replica Decoding Technique for Contention Resolution Diversity Slotted Aloha,” *IEEE 82nd Vehicular Technology Conference (VTC Fall)*, 2015. doi:10.1109/VTCFall.2015.7390798

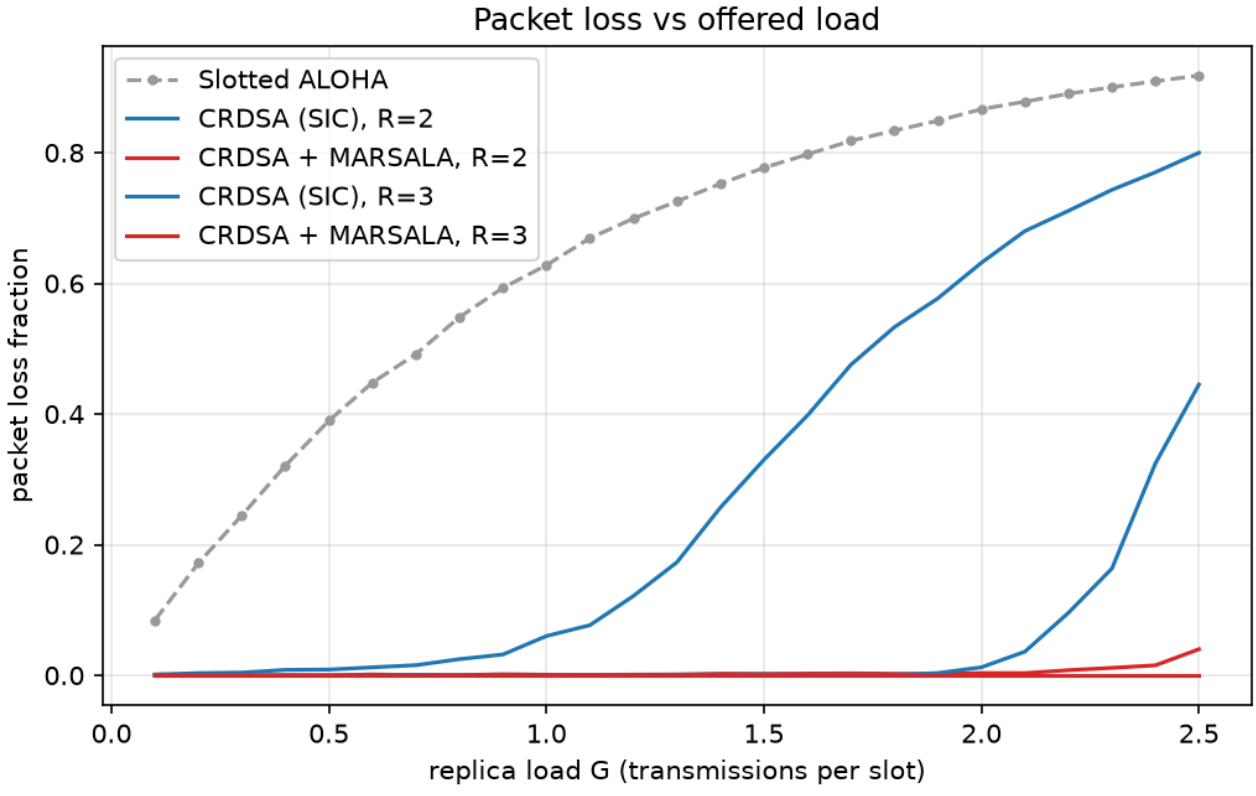
Results figures



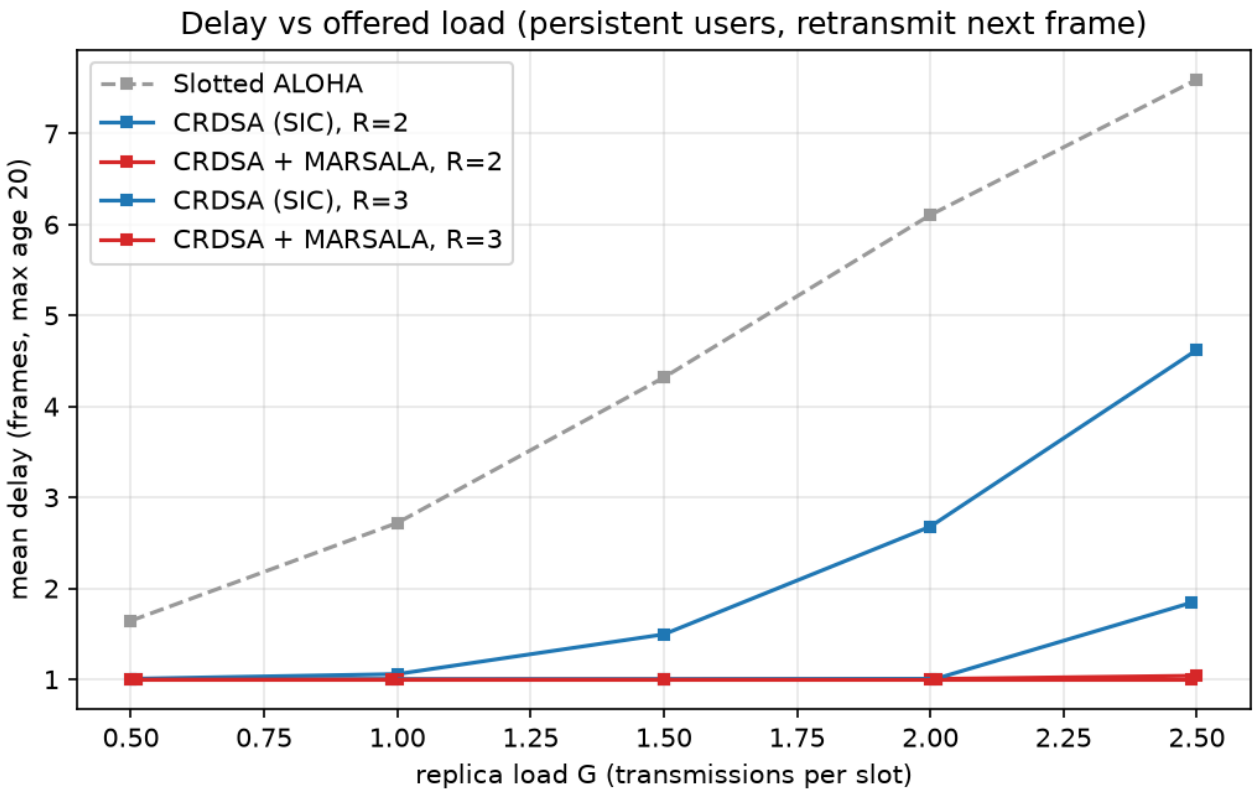
How SNR, decoding threshold, and replica count change delivery at equal user load



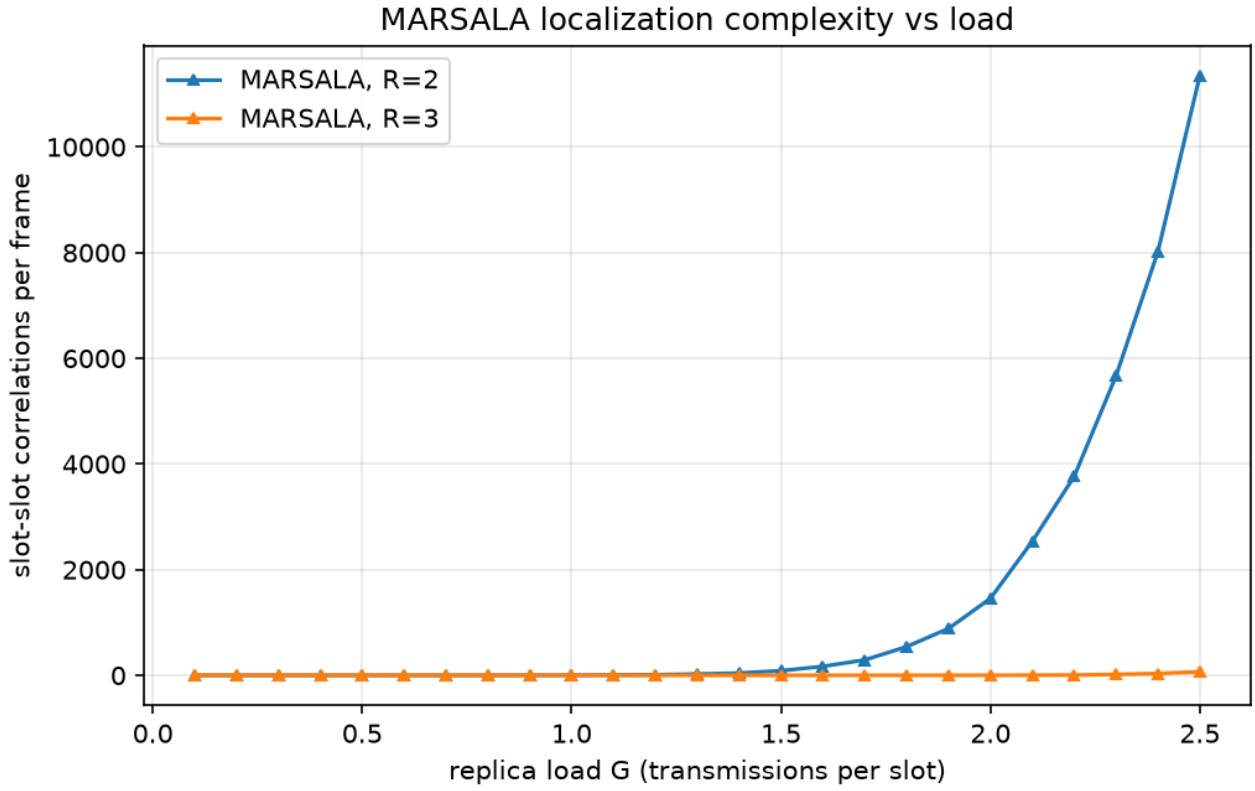
Throughput



Packet loss



Delay



Correlation work